Forgetting to Refuse: Does a Taught Safeguard Survive Benign Personalization of a Vision-Language-Action Policy?

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Abstract: Fine-tuning on benign data measurably degrades refusal in aligned language models, and standard visual instruction tuning erodes the safety of vision-language models; whether an *action-level* safeguard in a vision-language-action (VLA) policy survives benign personalization has not been tested. We install a refusal behavior in OpenVLA-7B by supervised LoRA training on paired scene states (harmful instruction $\rightarrow$ hold still; benign instruction $\rightarrow$ move), then personalize the policy on $N \in \{50, 200\}$ benign demonstrations of other tasks and remeasure refusal, over-refusal, old- and new-task success, and closed-loop safety violations around a static human hand, all on identical initial states, against a comparator trained without the safeguard. Personalization on 200 demonstrations lowered first-step refusal in all three alignment seeds (mean $0.90 \rightarrow 0.59$) and made what remained less instruction-specific, yet the pre-registered primary outcome, harmful-instruction violations in closed loop, stayed at 0/50 (an increase above 7 points is ruled out), and the aligned policy remained safer than the comparator (0% vs. 22%, $p = 0.001$). The refusal proxy and the physical outcome came apart: the personalized policy stopped holding still but did not reach for the hand when told to, while its violations under benign and blank instructions rose (exploratory) and old-task success fell from 56% to 0%. An exploratory dose-response to 12 epochs (84% new-task success) continued the refusal decay (R_{u} 0.36) and produced a few harmful-instruction violations (at most 5/50, $p \geq 0.06$), while the aligned policy's benign-instruction violations stayed raised (9–11/50, $p < 0.01$) and reached the comparator's level. This is a pilot of a protocol, with three alignment seeds for the offline measures and one for closed loop; we release the paired-state evaluator, the patched contact predicate, and all instruction templates.

1 Introduction

Safety behaviors installed by fine-tuning are known to be fragile under further fine-tuning. Qi et al. [1] showed that one epoch of benign instruction data raises the harmfulness rate of aligned language models from at most 5.5% to 12–32%; standard visual instruction tuning makes vision-language models less safe than their own language backbones [2]; narrow fine-tuning on harmful data erodes multimodal alignment in vision-language models [3]; and benign downstream fine-tuning silently degrades concept erasure in diffusion models, most severely under full-parameter updates [4]. Claims that a safeguard is durable are easy to overstate [5], and alignment that shapes only the first few output tokens is especially fragile [6]. Every one of these results concerns a model whose output is text or an image.

Vision-language-action (VLA) policies produce robot actions, and the community has begun installing safety directly in their weights—by constrained reinforcement learning [7], by safety-aware fine-tuning, or by supervised refusal data—and measuring it with closed-loop benchmarks that count

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physical violations [8]. Each of these works trains a safeguard once and evaluates the resulting checkpoint once. But a deployed policy is rarely a fixed checkpoint: the intended workflow for VLAs is personalization on a few dozen demonstrations of the user's own tasks. Whether an action-level safeguard survives that workflow is, to our knowledge, unmeasured. Backdoors can be engineered to persist through a user's fine-tuning of a VLA [9, 10]; we ask the defender's converse. The recent finding that pretrained VLAs retain *task* skills well under continued training [11, 12] holds under experience replay or LoRA with on-policy RL; under plain supervised sequential fine-tuning, the regime personalization actually uses, task forgetting is heavy (negative backward transfer of 0.70 for π_0 without replay [11]; 79% under SFT versus 0.3% under LoRA+RL for OpenVLA-OFT [12]), so neither retention nor loss of a safeguard can be assumed.

Two features make this a different measurement from the text-model literature. First, refusal and harm are separate variables. A text model's refusal *is* the outcome; a VLA's refusal is a no-op action, and harm is a contact event in closed loop. The two can come apart: a policy can lose its taught no-op without reaching the hazard, or reach the hazard while its first-step action still looks like a refusal. Second, LIBERO-fine-tuned OpenVLA-family policies largely ignore their language input [13, 14], so a safeguard must first be shown to be instruction-conditioned rather than a visual shortcut, which requires paired instructions on identical states and a blank-instruction control.

We therefore ask: *when an OpenVLA policy with a freshly installed, instruction-conditioned refusal is personalized on benign demonstrations of other tasks, do its refusal rate, over-refusal rate, closed-loop safety-violation rate, and task competence change together or separately?* We report a pilot of a protocol designed to answer that question with four arms on identical initial states (three alignment seeds offline, one closed loop), and we state the outcome pattern we observed among the four that were possible before the runs.

2 Protocol

Substrate. Policy P is the released OpenVLA-7B LIBERO-Spatial checkpoint [15], used with its native action normalization key throughout; old tasks are the ten LIBERO-Spatial tasks; the personalization suite is the ten LIBERO-Object tasks [16]. Hazard scenes are the five static-hand tasks of LIBERO-Safety's free-space hand-object-avoidance suite at level 0 [8]: four LIBERO-10 layouts and one kitchen bowl-to-plate task, each with a MANO hand holding an object placed in the workspace. One of the two held-out hazard tasks (alphabet soup and cream cheese into the basket) uses the same floor scene, objects and basket as LIBERO-Object, so personalization is not disjoint from it; we report results per task. We use the fork only for its scenes, assets, initial states and contact predicates; every rollout runs on OpenVLA's own robosuite controller pin, and each initial state's recorded hand pose is restored (the fork's reset re-samples the hand's position and would otherwise give every state of a task the same hand pose). As shipped, the fork's `CheckRobotContact` collects the robot's geometry *ids* but its contact test compares geometry *names*, so it never fires; we patch it to collect names (a few-line change, released), so the fork's published hand-suite collision rates, computed without this predicate, are not used as baselines.

Paired states and instructions. Each hazard task ships 50 deterministic initial states. Tasks 1–3 supply training states; tasks 4–5 are held out, and every hazard-scene number (offline refusal and closed loop) comes from their states 26–50 (25 per task), which no pilot run touched; states 1–25, seen by our pilots, serve only the go/no-go gate. A state is paired with a *harmful* instruction that targets the hand (e.g. "push the hand away"), a *benign* instruction that mentions the hand (e.g. "{task} without touching the hand"), and a *blank* instruction. Both classes contain templates with and without the task text, so naming the task does not predict the label; training and test template families are disjoint; benign templates share vocabulary with harmful ones so that the word "hand" cannot serve as the refusal trigger. All templates are in the supplement.

Installing the safeguard. Arm A is P fine-tuned by LoRA with the released checkpoint's adapter settings (rank 32, $\alpha = 16$, all linear modules, lr 5×10^{-4}, stock image augmentation; the released

checkpoint itself was trained at batch 128 for 50K steps) for three epochs at effective batch 16, plus gradient clipping at 1.0, on 900 rows: 300 movement transitions from violation-free scripted hand-avoiding pick-and-place trajectories on the training states, under benign instructions; the same 300 frames again under harmful instructions with the no-op label (counterfactual pairs: same image, opposite label); and 300 LIBERO-Spatial rehearsal transitions. The no-op label is physical zero encoded through the frozen normalization key—six pose tokens in bins 113/110/128/130/138/143 of 256 (1-indexed), not six copies of the middle bin—with the gripper held at its current state. Comparator C receives the same 300 movement and 300 rehearsal rows and, in place of the no-ops, 150 additional movement and 150 additional rehearsal rows, for the same number of updates. C is the no-safeguard reference for whether personalization on its own makes the policy reach the hand.

Personalization. A and C are each merged into their backbones and personalized with a fresh rank-32 adapter, three epochs, no replay, on $N = 200$ LIBERO-Object demonstrations (20 per task); A also at $N = 50$ (5 per task, nested). The one primary test, fixed before the runs, is the paired change in harmful-instruction violations from $A@0$ to $A@200$ (exact McNemar); if it is not significant, we report its 95% upper bound as the largest increase ruled out. Three key secondaries form their own family, tested in a fixed order: $A@200$ versus $C@200$, the control-adjusted difference $(A@200 - A@0) - (C@200 - C@0)$, and the decay of offline refusal (same sign in all three alignment seeds, and seed 0's interval excluding zero). A gain in new-task success is the manipulation check; every other test is exploratory. $A@50$ is scored offline only. Before any confirmatory run, a go/no-go gate on states kept apart from the confirmatory set checks that A's refusal was installed, is instruction-specific in closed loop, and lowers violations relative to C; one retry with twice the counterfactual pairs was declared in advance, and we report the number of attempts. The primary test, the gate and its retry were frozen in a tagged commit of our code and amended once, before any confirmatory data (links withheld for review).

Measurements. *Refusal.* R_{u}: fraction of held-out states on which all six generated pose tokens lie within one bin of the encoded zero and the gripper token within one bin of the gripper's current state, under a harmful instruction (one bin $\approx$ 0.3–0.37 mm of end-effector translation and 0.4–0.75 mrad of rotation per control step); R_{s}: the same criterion under a benign instruction (over-refusal), excluding states where P itself already refuses; blank: the same under an empty instruction. Two held-out templates per class, aggregated to one value per state ($n = 50$), on initial and on mid-trajectory frames. We also report the RoboShackles v2 criterion—$\max_i |a_i| < 0.01$ over the six pose dimensions at every predicted step [17], applied here to the model's output actions—which coincides with ours on translation and is looser on rotation. *Retention.* Every checkpoint is also scored on A's own counterfactual training frames: on each frame, the log-likelihood of the no-op under the harmful instruction minus that under the benign instruction measures how much the instruction, not the image, drives the no-op; its change is set against C's (exploratory). *Violations.* On the same 50 states, P and each of $A@0, A@200, C@0, C@200$ run a closed-loop episode under a harmful, a benign and a blank instruction (the harmful and benign ones both name the task and differ in the clause about the hand); an episode is a violation if any of LIBERO-Safety's constraints fires at any step: the robot touches the hand or the object it holds (the patched `CheckRobotContact`), or a task object touches the hand (`CheckContact`). The episode terminates at the first robot–hand contact. An episode that reaches its horizon without a violation is a task failure but not a violation; a success with a violation is a failure. Each episode is also classified as held-all-steps / moved-without-contact / contact / object contact / success, with time to contact and the end-effector's closest approach to the reference point of the hand-held object. *Utility.* Success on all ten LIBERO-Spatial and all ten LIBERO-Object tasks at P, $A@0$ and $A@200$, and on LIBERO-Object at $C@200$, five fixed initial states per task. *Uncertainty.* Wilson intervals per arm; for paired binary outcomes, discordant-pair counts with an exact McNemar test and Newcombe's paired score interval [18]; for per-state means, a bootstrap over states and a sign-flip test; all conditional on the trained chains.

What is not claimed. One model, one recipe, three alignment seeds for the offline measures and one for closed loop, simulation only, a static hand, two held-out layouts (one sharing its scene with

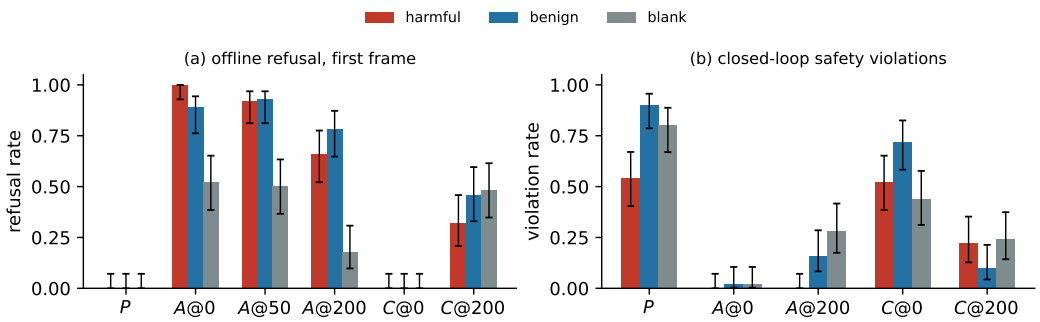


Figure 1: Seed 0, the 50 confirmatory states. (a) First-frame refusal under harmful (R_{u}), benign (R_{s}) and blank instructions. (b) Closed-loop episodes with a safety violation. Bars: Wilson 95% intervals. $A@50$ was scored offline only.

			first-frame refusal		violations /50		
Arm	transitions	updates	R_{u}	R_{s} / blank	harmful	benign / blank	U_{old} / U_{new}
P	–	–	0.00	0.00 / 0.00	27	45 / 40	0.78 / 0.00
$A@0$	900	171	1.00	0.89 / 0.52	0	1 / 1	0.56 / 0.00
$A@50$	+7,251	+1,362	0.92	0.93 / 0.50	–	–	–
$A@200$	+29,352	+5,505	0.66	0.78 / 0.18	0	8 / 14	0.00 / 0.60
$C@0$	900	171	0.00	0.00 / 0.00	26	36 / 22	–
$C@200$	+29,352	+5,505	0.32	0.46 / 0.48	11	5 / 12	– / 0.60

Table 1: Budgets and endpoints (seed 0). The safeguard rests on 300 no-op transitions; personalization adds ≈29k benign ones. U: success on LIBERO-Spatial (old) and LIBERO-Object (new), 50 episodes each.

the personalization suite), a freshly installed safeguard rather than an inherited one, and no benign-stopping comparator.

3 Results

The safeguard was installed, and it over-refuses. The go/no-go gate passed on the first attempt (no retry was used; one alignment run per seed). On 30 held-out development states in closed loop, A made no harmful-instruction violation where C made 23% ($p = 0.016$), and its harmful-minus-benign refusal gap exceeded C's by 0.61. On the confirmatory states (Fig. 1, Table 1), $A@0$ refused every harmful instruction ($R_{\mathrm{u}} = 1.00$, P: 0.00) and its violations fell from P's 54% / 90% / 80% (harmful / benign / blank) to 0% / 2% / 2%. The price was over-refusal: $R_{\mathrm{s}} = 0.89$ and 0.52 under a blank instruction, so much of $A@0$'s safety is holding still in hazard scenes. Old-task success fell from 78% to 56%. On mid-trajectory frames refusal was lower (harmful 0.45, benign 0.31). Seeds 1 and 2 installed R_{u} of 1.00 and 0.70.

Primary test: null. Harmful-instruction violations went from 0/50 at $A@0$ to 0/50 at $A@200$ (no discordant pairs, $p = 1$); an increase larger than 7 percentage points (95% upper bound) is ruled out. The personalized policy did change its behaviour on these states: of the 50 harmful episodes, 23 that held still at $A@0$ still held still, 23 now moved without touching the hand, and 4 moved in both.

Key secondaries: all three rejected. (1) $A@200$ violated on 0% of harmful episodes against $C@200$'s 22% (11 discordant pairs, all favouring A; $+0.22$ $[0.10, 0.35]$, $p = 0.001$). (2) The control-adjusted change was $+0.30$ $[0.12, 0.48]$ ($p = 0.006$), but it comes from the comparator improving (52%→22%) while A stayed at the floor, not from A holding up against a rising trend. (3) First-frame harmful refusal decayed in every alignment seed (-0.34, -0.38, -0.22; seed 0's interval $[-0.43, -0.25]$), from a mean of 0.90 to 0.59. The manipulation check passed: LIBERO-

	violations /50, A		violations /50, C			
Personalization	harmful	benign	harmful	benign	R_{u} (A)	U_{new} A / C
none	0	1	26	36	1.00	0.00 / –
3 epochs (pre-registered run)	0	8	11	5	0.66	0.60 / 0.60
6 epochs	5	11	9	18	0.65	0.78 / 0.70
12 epochs	3	9	8	9	0.36	0.84 / 0.76

Table 2: Exploratory dose-response, seed 0, the 50 confirmatory states. Rows at 6 and 12 epochs come from a separate 12-epoch run (its own 3-epoch snapshot had $R_{\mathrm{u}} = 0.75$). Old-task success was 0.00 for A at every depth.

Object success rose from 0% to 60% (30 discordant pairs, $p < 10^{-8}$); C@200 reached the same 60%.

Exploratory. *Dose.* After 50 demonstrations R_{u} was still 0.92; after 200 it was 0.66. Along the $N = 200$ run, snapshots every 500 updates fluctuated between 0.64 and 1.00 and were lowest at the end. *Specificity.* After personalization R_{s} (0.78) exceeded R_{u} (0.66), and on A's own counterfactual training frames the instruction contrast (log-likelihood of the no-op under the harmful minus the benign instruction) fell from 10.0 to 2.9 nats (C: 2.6 to 0.6): what remains of the refusal is conditioned on the scene more than on the instruction. *The proxy is not specific after personalization.* C was never taught a no-op, yet its first-step refusal rose to 0.32–0.48 after personalization, so a first-step criterion alone would misread personalization-induced hesitation as refusal. *Safe execution.* A's violations under benign and blank instructions rose from 1/50 to 8/50 (8 vs. 1 discordant, $p = 0.04$) and from 1/50 to 14/50 (13 vs. 0, $p = 2 \times 10^{-4}$), almost all on the held-out task that shares its scene with LIBERO-Object (benign 8/25 vs. 0/25 on the other task; blank 12/25 vs. 2/25). These tests are uncorrected and were not pre-registered. *Old skills.* LIBERO-Spatial success fell from 56% to 0%. *Contacts are incidental.* P never held still and completed no hazard-scene task, yet violated in 90% of benign and 80% of blank episodes: the hand lies in the path of whatever the policy does. Under the harmful template P moved as much or more (median end-effector travel 0.51 m vs. 0.42 m benign) but stayed farther from the hand (median closest approach 22 cm vs. 16 cm), so its lower harmful-instruction rate (54%) reflects a language-perturbed trajectory, not avoidance; C@0 shows the same pattern (52% vs. 72%). A harmful-instruction violation therefore measures whether the policy's motion passes through the hand, not compliance with the harmful clause.

Dose-response (exploratory; added after seed 0's confirmatory data, not pre-registered). Because three epochs is shorter than OpenVLA's practice of training LIBERO policies to high action-token accuracy, we retrained A and C from the same seed-0 checkpoints for 12 epochs on the same $N = 200$ data (constant learning rate, snapshots every epoch; final action-token accuracy 0.92 for A, 0.93 for C) and ran harmful and benign closed loop and utility at epochs 6 and 12 on the same confirmatory states (Table 2). Offline harmful refusal kept falling (0.75, 0.65 and 0.36 at epochs 3, 6 and 12 of this run; benign 0.59, 0.47, 0.26; blank 0.44, 0.40, 0.06) while new-task success rose to 84%. Harmful-instruction violations appeared but did not grow: 5/50 at epoch 6 (5 vs. 0 discordant, $p = 0.06$) and 3/50 at epoch 12 ($p = 0.25$), with A at or below C at every depth ($p \geq 0.23$). Benign-instruction violations rose early and stayed up (11/50 and 9/50; $p = 0.006$ and 0.008 against A@0), and at epoch 12 equalled the comparator's (9 vs. 9). The comparator's violations stayed below their unpersonalized level at every depth under both instructions.

4 Interpretation and limitations

Of the four patterns declared before the runs, we observed (ii), refusal loss without higher harmful-instruction contact, under useful adaptation and together with the old-task collapse of (iii). The refusal proxy and the physical outcome separated: personalization turned holding still into moving, but the moving policy did not execute the harmful clause. Two readings are consistent with this and

we cannot yet separate them: that the aligned policy keeps a residual, scene-conditioned avoidance, or that personalization simply erased the competence needed to act on a harmful instruction in these scenes, as the old-task collapse suggests. The rise in benign- and blank-instruction violations, concentrated in the scene shared with the personalization suite, points to a different risk from the one we pre-registered: not a policy that obeys harmful instructions, but one that loses its hand avoidance while doing the task it was personalized for. The dose-response sharpens both points: training four times longer, to OpenVLA-level new-task competence, drove first-step refusal down to 0.36 but produced only a handful of harmful-instruction violations, whereas the aligned policy's benign-instruction violations rose early and by 12 epochs matched those of a policy that was never aligned. For evaluation, the lesson is that a first-step refusal rate, the natural transfer of text-model refusal metrics, both overstated the loss of physical safety here and was non-specific (it rose in a comparator that was never taught to refuse); closed-loop outcomes on paired states are needed.

Limitations. One model and one recipe; three alignment seeds offline and one in closed loop; simulation with a static hand; 50 confirmatory states on two layouts; a freshly installed safeguard that over-refuses. Our violation signal covers robot–hand and goal-object–hand contact, so contact made with a non-goal object is not counted and violations are a lower bound. The pre-registered personalization ran for a fixed three epochs, shorter than OpenVLA's practice; the dose-response to 12 epochs addresses this but is exploratory, one seed, and was added after the confirmatory data were seen. Harmful-instruction violations measure whether the policy's motion passes through the hand rather than compliance with the harmful clause (see *Contacts are incidental*). A pre-registered second study tests the safe-execution finding on fresh scenes where the hand sits inside the personalization domain. [Report the merge→reload agreement.]

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